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Chapter 7. Exit nodes, the Mullvad add-on, and what I actually run

The payoff of this chapter: the one rule that governs VPNs on the road, the honest setup I actually live with (a standalone Mullvad I switch on and off, giving up homelab access while it's on), what a Tailscale **exit node** is, and the paid add-on that *can* give you privacy and your homelab at the same time, if you're willing to trade control and a few more dollars for it.

At home, everything is easy: your devices share the Pi's network and everything just works. The moment you walk out the door, the only thing connecting your phone to your Pi is **Tailscale**. This part is the field manual for that world, and it picks up right where [Chapter 6](#) left off: you've chosen *which* VPN you want (Mullvad, for its openness); now we deal with how it actually coexists with the homelab.

Reaching your homelab from anywhere (the easy part)

This one already works, and it's the cleanest win. Because Tailscale is a mesh VPN, any device signed into your tailnet reaches the Pi from anywhere, coffee shop, cellular, hotel, with no extra setup:

- **Audiobookshelf:** <https://abs.home> streams just as it does at home.
- **The dashboard:** <https://home.home> (or <https://homelab>) opens from anywhere.

The reason it's effortless is that you're only reaching *into* your own network over an authenticated tunnel. The hard parts below are all about the *outbound* direction, what your traffic does and which DNS resolves it.

Pi-hole already follows you (set up in Chapter 4)

You don't need to do anything here, when you made Pi-hole your tailnet's DNS in [Chapter 4](#) (the **Override local DNS** setup), you also got Pi-hole's ad-blocking on every device, everywhere. On cellular, your phone still resolves through Pi-hole, so ads stay blocked. The same step is also what makes your `.home` names work away from home.

That's the good news. The catch is what happens when you turn on *another* VPN, which is the rest of this chapter.

! This only holds while Tailscale is the active VPN

The whole mechanism rides on Tailscale being the thing controlling your device's DNS. The instant another VPN takes over (next section), this stops applying. That's not a bug, it's the central tension of going mobile, and the rest of this chapter is about navigating it.

The one rule that governs everything off-network

Here it is, the rule that explains every “why can't I just...” on the road:

A full-device VPN owns the default route *and* the DNS, and a device runs only one such VPN at a time.

Two consequences fall out of it:

1. **A privacy VPN and Tailscale can't both be the active tunnel.** Phones make this a hard wall: iOS and Android allow exactly one active VPN, so turning on Mullvad (or Aura) drops Tailscale. But, and this surprised me, **the same thing bites on a laptop in practice.** A Mullvad WireGuard tunnel grabs the entire default route, which swallows the path back to your tailnet, so the moment Mullvad is up, `https://homelab` and your `100.x` addresses go dark there too. The phone enforces “one VPN” by policy; the laptop enforces it by routing.
2. **Whatever VPN is active forces its own DNS.** Even setting the one-VPN limit aside, the active VPN routes DNS through its own resolvers. So while you're on *any* location/privacy VPN, your traffic is *not* going through Pi-hole, by design, to prevent DNS leaks.

Insight This is why “use my own VPN *and* filter through Pi-hole *and* reach my homelab, all at once, on one device” has no clean answer with two separate apps: Pi-hole and your homelab live only on your tailnet, reaching them needs Tailscale to be the active tunnel, and a privacy VPN that's up takes that role (by policy on a phone, by routing on a laptop). The only real escapes are to put the privacy VPN and your homelab on the *same tunnel*, which is the exit-node idea further down, or to simply switch between them, which is what I do.

What I actually run: keep them separate, switch per activity

After trying to have it all at once and losing (see the sidebar), I settled on the setup that's boring and bulletproof: **I keep a standalone Mullvad subscription and Tailscale installed side by side, and I only ever run one at a time.** On both my laptop and my phone.

In practice that means I segregate my activities into two modes and flip between them in a couple of seconds:

- **Tailscale on, Mullvad off (my default).** Homelab reachable, Pi-hole ad-blocking everywhere via the DNS push, `.home` names resolve. This is where I live almost all the time. The cost is that my real IP is showing.
- **Mullvad on, Tailscale off (when privacy matters).** Sketchy Wi-Fi, or I want to change region or hide my IP. While it's on I accept that the homelab and Pi-hole are simply *gone* for that stretch, Mullvad carries my DNS and its own ad-blocking instead. When I'm done, Mullvad off, Tailscale back on.

That's the whole system. No exit node, no policy routing, no add-on. The honest downside is right there in the open: **I can't reach my homelab while the privacy VPN is on**, so I plan around it (do the private thing, then switch back to grab something off the Pi). For me that beats the alternatives, because I keep full control of my own Mullvad subscription, its config files, its settings, and I'm not paying for anything extra.

i I tried hard to have both on one box, and couldn't make it stick

Before settling on switch-per-activity, I spent real hours trying to run Mullvad and Tailscale *simultaneously* on my laptop. The approaches, and why each fell over:

- **Split tunneling / carving out the tailnet.** Mullvad's WireGuard config claims the entire default route (`AllowedIPs = 0.0.0.0/0`), which also swallows the return path to Tailscale's range (`100.64.0.0/10`). The plan was to exclude that range from the Mullvad tunnel so my devices could still reach the homelab.
- **Forcing Tailscale's route priority** so its `100.x` routes would win over Mullvad's catch-all default.

I could occasionally get a fragile version working, but never one that survived a reconnect, a Mullvad server change, or a reboot. It was never reliable enough to trust. The lesson I took: with two separate full-device VPN apps, having both *truly* on at once is a fight you'll keep losing. If you want both at the same time and reliably, you don't fight the routing, you change the architecture, and that's the exit-node add-on below.

The thing that *would* let you have both: an exit node

Everything above assumes two separate VPN apps. There's a different architecture that sidesteps the whole "one tunnel at a time" problem, and it's worth understanding even if (like me) you don't use it yet.

Normal Tailscale is an **overlay network**: it only carries traffic *between* your own devices (laptop homelab phone). Your ordinary internet traffic still leaves through your local connection; Tailscale doesn't touch it.

An **exit node** changes that. You designate one device on your tailnet as “the exit,” and your other devices route **all** their internet traffic through it, the full default route, not just tailnet traffic. To the outside world, your traffic now appears to come from the exit node's IP. That's exactly how a traditional full-tunnel VPN behaves, except the “VPN server” is a node on your *own* tailnet, so **you're still on Tailscale the whole time**. Only one exit node is active per device, and you switch it off by selecting “None”:

```
tailscale exit-node list           # list available exit nodes
tailscale set --exit-node=<node>  # route everything through it
tailscale set --exit-node=        # turn it back off
```

The key insight: because an exit node *is* still Tailscale, choosing one doesn't drop your tailnet. Your homelab stays reachable while your traffic exits somewhere else. That's the one architecture that escapes the one-rule trap, and it comes in three flavors.

Flavor 1 (the upgrade path): the Tailscale Mullvad add-on

This is the option that buys you privacy *and* your homelab at the same time, and it's the thing I'd reach for if my needs grow. Tailscale sells access to **Mullvad's worldwide WireGuard servers as a paid add-on purchased through Tailscale**. Mullvad's servers simply appear as selectable exit nodes inside your tailnet:

```
tailscale exit-node list           # Mullvad cities now appear here
tailscale set --exit-node=<mullvad-node> # exit through Mullvad, stay on
Tailscale
```

On **iOS / macOS / Android**: the **...** menu → **Use exit node** → pick a Mullvad **location**. Because it's still Tailscale, your homelab and **.home** names keep working while your traffic exits through Mullvad. That's the genuine best of both worlds: always-on privacy that doesn't cost you the homelab, and it's the only option that works cleanly on a phone without fighting the one-VPN rule.

So why don't I run it? Because of what you give up:

- **Cost:** about **\$5/month** on top of things. If you also keep your own Mullvad subscription (as I do, for the control), you're now at **\$10+/month** for the pair. The add-on alone (~\$5) can *replace* your own sub, but then you lose the next point entirely.
- **Control.** When you buy the add-on, **Tailscale provisions a Mullvad account for you that you have very little say over**. You don't get your own Mullvad login, you can't download WireGuard config files to run on the Pi or a VPS, and you can't tune Mullvad's own settings. You get exit nodes and nothing else. For someone who specifically wants to *own and control* their Mullvad config (which is the whole reason Chapter 6 pointed at Mullvad), that's a real loss.

Insight This is the honest trade at the center of the chapter. The add-on gives you *convenience* (one tunnel, always on, homelab intact). Owning your own Mullvad sub gives you *control* (portable configs, full settings, no extra account you don't manage). Right now I value control more, and I'm fine switching apps by hand to get it. If my needs shift toward "always-on privacy that never costs me the homelab," I'll pay the extra and run both, or move fully to the add-on. It's a budget-and-priorities call, not a right-or-wrong one.

When you *do* use a Mullvad exit node, two flags matter on the road:

```
# Keep access to LAN devices (printers, etc.) while using an exit node
tailscale set --exit-node=<mullvad-node> --exit-node-allow-lan-access
```

- **--exit-node-allow-lan-access:** by default, turning on *any* exit node cuts your access to the local network you're physically on. This flag restores it. On mobile it's the **Allow LAN access** toggle when you pick the exit node.
- **DNS leaks:** allowing LAN access can let some DNS queries escape the tunnel. Recent Tailscale clients (1.48.3+) route DNS correctly through Mullvad exit nodes without extra config, so keep your clients updated.

 A real bug to watch for

Some Tailscale client versions around 1.82 had a defect where an **active exit node bypassed the global Pi-hole nameserver**, so your ad-blocking would quietly stop whenever an exit node was on. If you see that symptom, update the Tailscale client first. It's a known issue, not your misconfiguration.

Flavor 2: your own Pi as an exit node (free, but not private)

You can advertise one of your *own* machines as an exit node. The helper script in this chapter's folder does it on the Pi:

```
sudo tailscale set --advertise-exit-node
# then approve it: admin console -> Machines -> the Pi ->
#   Edit route settings -> enable "Use as exit node"
```

On Linux you also enable IP forwarding (`net.ipv4.ip_forward=1` and the IPv6 equivalent); the script sets both.

- **What it's good for:** routing through your **home** Pi so you appear to be at home, reaching home-only services, or content tied to your home connection, and, importantly, it's the one exit option that keeps your traffic on your own network where **Pi-hole still applies**. This is the sweet spot when what you want on the road is ad-blocking, not IP-hiding.
- **What it's not:** privacy. Traffic exits from **your own home IP**, so there's no hiding. And don't use a *phone* as an exit node, it routes in userspace and is slow; the Pi (kernel routing) is fine.

Flavor 3 (advanced): a Pi-style exit node chained through Mullvad

Flavor 2 hides nothing, because the box exits from your own IP. You can fix that by *chaining*: run an exit node that itself tunnels out through a standalone Mullvad **WireGuard** config. Your devices reach the node over the tailnet, and the node forwards their traffic out through Mullvad:

you -> Tailscale -> your exit-node box -> Mullvad WireGuard -> internet

This is the only way to use a **standalone** Mullvad subscription *as a Tailscale exit node* without paying for the add-on, so it keeps the control you wanted while still giving you the always-on-plus-homelab shape. The shape of it:

1. Use a machine that is **not your home Pi** if privacy is the goal (a cheap Linux VPS works well). The home Pi only makes sense for “appear at home,” where adding Mullvad would defeat the purpose.
2. On that box, install Tailscale, advertise it as an exit node, and enable IP forwarding (same as Flavor 2).
3. Download a **WireGuard config** from your Mullvad account and bring it up (`wg-quick up <conf>`) so the box’s default route leaves through Mullvad.
4. The sharp edge: Mullvad’s config claims the **entire** default route (`AllowedIPs = 0.0.0.0/0`), which also swallows the return path to your tailnet. You have to keep Tailscale’s range (`100.64.0.0/10`) out of the Mullvad tunnel so your devices can still reach the node. That policy-routing fiddliness is exactly the same fight I lost on the laptop, just moved to a box you can leave running, and it’s the work the add-on saves you.

Reach for this if you already pay for Mullvad, want to keep your own config, and enjoy the plumbing. For most people the add-on (Flavor 1) buys the same outcome with none of the routing surgery.

The decision matrix

You can’t have all three of {homelab access, Pi-hole filtering, privacy VPN} on one device at one time with two separate apps. But you *can* switch between coherent modes in seconds. Pick per situation:

You’re away and you want...	Set this	Pi-hole?	Homelab?	Hides IP?
Homelab + ad-blocking (my everyday default)	Tailscale on, no exit node	via DNS push		(your real IP)
Privacy, my way (what I run)	Tailscale off, standalone Mullvad on	Mullvad’s DNS		
Appear at home / reach LAN	Tailscale + Pi as exit node (Flavor 2)			(home IP)

You're away and you want...	Set this	Pi-hole?	Homelab?	Hides IP?
Privacy <i>and</i> homelab at once	Mullvad add-on exit node (Flavor 1)	Mullvad's DNS		

The pattern to read out of that table: with two separate apps (rows 1 and 2) you trade the whole homelab for privacy and switch between them, which is cheap, fully under your control, and what I do. The exit-node rows (3 and 4) are the only ways to keep the homelab *while* your traffic exits elsewhere, and the only one of those that also hides your IP is the paid Mullvad add-on.

A practical everyday routine

For most people the comfortable default is simple, and it's mine:

1. **Leave Tailscale on, no exit node, all the time.** Homelab reachable, Pi-hole ad-blocking everywhere via the DNS push, phone behaves normally. You give up IP-hiding, which most of the time you don't need.
2. **When you specifically want privacy,** switch to standalone Mullvad for that session and accept that the homelab steps aside until you switch back. Cheap, reliable, fully yours.
3. **If “always-on privacy without ever losing the homelab” becomes worth the money and the loss of control,** buy the Mullvad add-on (Flavor 1) and let a Mullvad exit node do both at once, or stand up a chained exit node (Flavor 3) if you'd rather keep your own config.

Recap: and the whole series in one breath

- **Reaching your homelab away** is the free win: Tailscale makes `homelab:13378` and `https://home.home` work from anywhere, no extra setup.
- **Pi-hole on the road** rides on the Chapter 4 DNS push, and works only while Tailscale is your active VPN.
- **The governing rule:** a full-device VPN owns the route and DNS, and a device runs one at a time (by policy on a phone, by routing on a laptop), so a separate privacy VPN and your homelab can't both be live with two apps.
- **What I run:** a standalone Mullvad I keep fully under my own control, switched on only when I want privacy, accepting that the homelab is offline while it's on. Boring, reliable, cheap.
- **The escape hatch, when you want both at once:** an **exit node**, most easily the paid **Mullvad add-on**, which trades some control and a few more dollars for always-on privacy that never costs you the homelab.

With that, your homelab is fully mobile: an always-on Raspberry Pi that streams your media, blocks ads on every device, presents a single `https://home.home` control panel, and gives you a deliberate, switchable answer for privacy on the road. All of it private by default, over one Tailscale network, with nothing exposed to the public internet.

That completes the core of the homelab. **Volume III** is the extras: getting your *phone and your Linux machines* to talk to each other for everyday remoting, file transfer, and clipboard sharing, starting with [Chapter 8](#).